

MULTILABEL CLASSIFICATION OF UAV IMAGES WITH CONVOLUTIONAL NEURAL NETWORKS

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ABSTRACT

In this paper, we present a multilabel classification method for images acquired by means of Unmanned Aerial Vehicles (UAV) over urban areas. Due to the fact that UAV-grabbed images are characterized by extremely high spatial resolution, usual recognition schemes (such as traditional satellite or airborne based images) are likely to fail. In this work, tile-based multilabel classification framework is adopted to overcome such issue. In particular, a given UAV-shot image is first subdivided into a grid of equal tiles. Next, deep neural network-induced features are extracted from each tile and then fed into a radial basis function neural network classifier in order to infer the corresponding object list. We apply a refinement step at the top of the complete deep network architecture to boost the classification results. The proposed method was evaluated on a dataset acquired over the city of Trento, Italy with an hexacopter UAV. Superior classification rates have been scored with respect to the state-of-the-art.

Index Terms—unmanned aerial vehicle, extremely high resolution, image multilabeling, multi-object detection, image analysis, coarse description.

1. INTRODUCTION

The ever increasing interest witnessed in the acquisition and development of Unmanned Aerial Vehicles (UAVs), commonly known as drones in the past few years, has paved the way to a very promising and effective technology. Since 2005, the number of countries that have acquired drones doubled from 40 to more than 75 [1]. UAVs have proven their effectiveness in collecting data in unreachable areas and limited coverage zones due to their small size and fast deployment. Moreover, their custom-made capacity allows them to collect information with a very high level of detail, leading to extremely high resolution (EHR) images. UAVs were mainly created for military usage. However, in the last decade, they have been exploited in several civilian applications as well. For instance, in [2], a real time algorithm is introduced for classification, object detection and tracking from thermal UAV images acquired over the surface of the ocean. In [3], the authors present a UAV cloud system disaster surveillance system, to reduce natural or man-

made damages. In [4], Shaodan *et al.* introduce an unsupervised classification method for UAVs images to detect earthquake triggered on rural houses. Furthermore, several works dealing with UAVs vehicles detection can be found in [5]-[6].

In spite of the efforts being dedicated to UAV imagery classification and analysis within the remote sensing community, there is still a plenty of room for improvement. Indeed, as the spatial resolution increases, so does the need for new methods to process images with such high level of detail and rich information content, where traditional ways of classification such as pixel-based and segment-based descriptors may raise the problem of intra-class variability especially when dealing with several classes at the same time. Moreover, they dramatically increase the computational needs. Indeed, this makes the analysis of UAV imagery particularly challenging. In this work, we deal with the problem of multilabel classification of EHR images acquired by means of UAVs using a coarse description approach. That is, instead of attributing a label to each individual spatial entity or segment region descriptor as in the traditional monolabel classification, we describe the considered entity by a list of object classes present in it. This method subdivides the image into a grid of equal tiles, then using some specific tile representation and an opportune classification tool, we attribute to each tile a vector of labels representing the object classes that are possibly present.

In [7], Moranduzzo *et al.* presented a novel multiclass method for coarsely describing UAV images over urban areas. Their method derives benefits from exploiting Local Binary Pattern (LBP) feature descriptor. Their approach yielded interesting results. Recently, the computer vision community has reported a very promising generation of neural networks, called Convolutional Neural Networks (CNN) [8]. They show that the CNN can overcome traditional classification methods in very complex vision [9]-[10].

This paper aims at investigating the effectiveness of CNN, as a feature extractor for our multilabel classification problem in UAV images. In Section 2, we expose the proposed method to perform the classification by resorting to a Radial Basis Function Neural Networks (RBFNN) [11], as a multilabel classifier. While in Section 3, we present the experimental results obtained on real UAV images acquired over urban

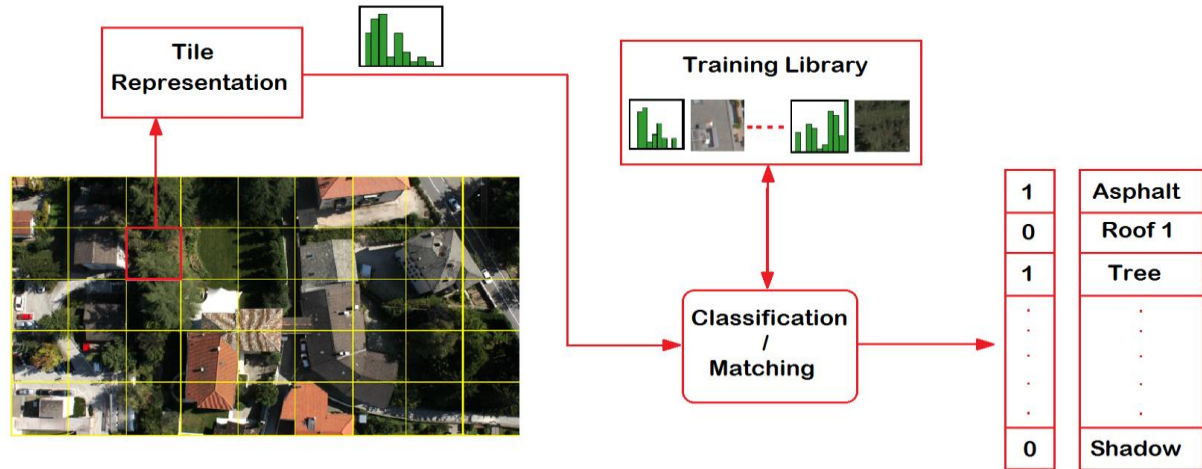


Fig. 1. Flow chart of the multilabel classification method.

areas and we compare them with state-of-the-art-results. A simple refinement step on the RBFNN outputs is implemented to improve the obtained outcomes.

2. PROPOSED METHOD

Let us consider a three-channel RGB extremely high resolution (EHR) image (I) acquired by means of an UAV. We subdivide it into a grid of tiles of equal sizes $I(x, y)$ (where (x, y) represent the pixel coordinates in image I). The size of each tile is defined according to the spatial resolution of the image (I) and the expected sizes of objects that we want to recognize.

The multiclass tile-based approach is based on two main stages:

1) *a suitable tile representation strategy*; and 2) *a tile classification/matching method*. A query tile $I(x, y)$ is labeled either with a binary vector of the most similar tile present in the training library using a matching strategy, where the closest tile in the feature space from the training library has likely the same list of objects of the query tile, or it is labeled by means of a classification paradigm (Fig. 1).

In particular, in this work, our method starts with the extraction of features using the GoogLeNet pre-trained CNN [8]. CNN is a feed-forward hierarchical neural network implementing a set of convolutional and subsampling operations, followed by a softmax classifier. The main underlying idea behind convolutional neural network, is to look for a pattern supposed to be invariant to spatial translations. The major difference between convolutional networks and a Multi-Layer Perceptron (MLP) is that, unlike MLP, the internal order of inputs and hidden units is relevant in convolutional networks, and each unit is connected with a specific spatial position of the input image. CNN architecture starts with the convolutional layer that is usually a 3 dimensional volume of neurons. It consists of a set of feature detectors which refer to a convolution with a mask that is consistently convolved across the width and height of the

input layer in order to extract very robust features against noise and translation. The repetition of this mask all over the input layer allows to share the same weights all over the input layer which reduces the number of free parameters learned. In fact, this simplifies the computational requirements of training the network on large datasets building much efficient and powerful networks. After the convolutional layer, comes the subsampling layer also called pooling layer. It is a nonlinear down sampling layer that reduces the spatial size thus the number of parameters to be computed for the next layer. The most common pooling technique is Max-Pooling. It divides the input into a set of non-overlapping blocks, and assigns to each block its maximum value.

In order to increase the sparseness of the decision function, an elementwise activation function Rectified Linear Units (ReLU) layer can be applied after the convolutional layers. The ReLU layer deals also with the vanishing gradient problem in the backpropagation. Thereafter a set of convolutional and subsampling layers, comes the classification layer. It is a fully connected layer that has full connections to all activation units in the previous layers with a loss function (e.g., softmax).

Training a deep network usually requires a huge number of training images to avoid overfitting. Nevertheless, one may reasonably tackle this issue by transfer learning, which in our case consists of exploiting the weights of a model that is pertained on a large dataset and making use of them while developing our dataset-specific classifier. CNN features computed with pre-trained networks have been used in many computer vision tasks, and has shown good results [12]-[13].

One of the publicly available pre-trained CNN is GoogLeNet. It was trained over ILSVRC2014 dataset, which contains over 1.2 million images, with a classification challenge of 1000 different classes, thus making GoogLeNet a promising candidate for generating powerful discrimination features. GoogLeNet is a 22-layers deep network excluding pooling

Table 1. Classification accuracy in terms of sensitivity (SENS) and specificity (SPEC) obtained by the CNN with Chi-squared matching distance, CNN-RBFNN, and refined CNN-RBFNN classifiers.

	SENS (%)	SPEC(%)	AVERAGE (%)
LBP-Chi_squared (χ^2)	61.91	89.32	75.61
CNN-RBFNN	70.60	94.31	82.45
CNN-RBFNN Refined	87.43	83.56	85.50

layers, with a softmax loss layer as a classifier. The size of its receptive field is 224×224 of three channels (RGB) with zero mean. A rectified linear activation (ReLU) is used in all its convolutional layers. GoogLeNet generates a feature vector of size equal to 1024.

Since GoogLeNet is not directly adapted to multilabel classification tasks, in this work, we substitute the linear softmax classifier with a Radial Basis Function neural network (RBFNN), which is a classifier that fits our multilabeling requirement. Indeed, the RBFNN can handle simultaneously multiple outputs, which may characterize each tile of the image. Each tile will be labeled with a binary vector of multiple predefined classes which correspond to the number of RBFNN outputs. Moreover, we will implement a simple refining mechanism at the output of the RBFNN so that to further boost the multilabeling accuracy. It is based on a thresholding operation applied on the output of the RBFNN. The value of the threshold is estimated empirically, i.e., by maximizing the average accuracy on the training samples.

3. EXPERIMENTAL RESULTS

In this section, we evaluate the classification capability of the proposed method on a real dataset of images acquired over the Faculty of Science of the University of Trento (Italy) on the 3rd October 2011 at 12:00 am. Acquisitions were performed with a picture camera Canon EOS 550D characterized by a CMOS APS-C sensor with 18 megapixels. The images are characterized by three channels (RGB) with a spatial resolution of approximately 2 cm. The obtained images are of size 5184×3456 pixels with 8 bits radiometric resolution. Our dataset is composed by 5 images, subdivided into two groups.

Training set: we selected 2 images as training images. We extracted randomly 1000 tiles of size 224×224 from both images. We rotated each tile randomly from one of these four angle values: 0° , 90° , 180° , and 270° . Given the image resolution of 2 cm, the tile size covers 4.5×4.5 meters. The two training images were chosen from the overall set since they contain most of our predefined classes of objects, which are: 'Asphalt', 'Grass', 'Tree', 'Vineyard', 'Car', 'Roof 1', 'Roof 2' and 'Shadow'. We have divided the roof class into two classes, due to its variability.

Test set: it is composed of 3 images. We subdivided the image test into non overlapping grid of equal tiles of 224×224 as explained in the coarse approach description. For the accuracy evaluation we used the sensitivity and specificity evaluation metrics in order to compare our method performance on this data set with the reference method.

As listed in Table 1, the results obtained from the combination of the GoogLeNet features and the RBFNN classifier achieves 70.60% of sensitivity and 94.31% of specificity. This combination overcomes the state-of-the-art method [7] on the same data-set. Using LBP extracted features with the chi-squared measure distance (χ^2) as a matching strategy achieves 61.91% and 89.32% of sensitivity and specificity, respectively, where each test tile is labeled with the same binary vector of the most similar tile present in the training library. As shown, the proposed method resulted superior accuracy values in both terms of sensitivity and specificity against the state-of-the-art method [7]. The experimental results sensitivity and specificity accuracies of each class are shown in Table 2. As shown, all the specificity and sensitivity results obtained by CNN-RBFNN approach surpasses the work reference in terms of both sensitivity and specificity. This points out that the CNN-RBFNN method has successfully detected with a high performance the 8 classes.

Table 2. Sensitivity (SENS) and specificity (SPEC) accuracy for all classes achieved by the CNN with Chi-squared matching distance, CNN-RBFNN, and refined CNN-RBFNN classifiers

		Asphalt	Grass	Tree	Car	Roof1	Roof2	Vineyard	Shadows	Total
LBP-Chi_squared (χ^2)	SENS	51.42	75.70	66.67	22.11	11.76	58.23	73.55	61.11	61.91
	SPEC	88.56	68.19	80.89	95.04	98.03	89.69	94.24	89.42	89.32
CNN-RBFNN	SENS	59.57	85.01	67.44	53.15	41.66	60.95	76.78	61.76	70.60
	SPEC	94.95	77.73	88.47	99.24	99.21	95.27	98.42	93.69	94.31
CNN-RBFNN Refined	SENS	81.56	94.86	84.16	78.37	75	82.85	91.66	85.29	87.43
	SPEC	85.12	56.02	63.11	93.93	96.48	84.96	91.70	82.00	83.56

A further refinement of the results has been possible thanks to the addition of a layer of step functions at the top of the RBFNN. The parameter (i.e., threshold) of these functions is estimated empirically by maximizing the average accuracy on the training samples. The results on the test tiles are reported in Table 1. As can be seen, a significant boost of accuracy was possible by thresholding the RBFNN outputs. It is interesting to observe that there is a clear average accuracy improvement that comes at the cost of some loss in the specificity. Despite the decrease of specificity of roughly 11 % from 94.31% to 83.56%, there is a higher gain of almost 17 % in terms of sensitivity. This refinement layer exhibits the advantage that it lowers the risk of missing objects resulting a better prediction method.

4. CONCLUSION

In this work, we have presented a multilabel classification method for UAV images. The proposed method relies on subdividing the image into a set of equal tiles, where each tile is represented by means of a vector of labels that lists the ensemble of objects present in it. CNN features are extracted from all the tiles using the GoogLeNet pretrained network. Then a RBFNN model is trained for the multilabel classification task. A simple thresholding refinement step has been applied on the RBFNN outputs to enhance the obtained results. From the results, one can infer that our method is rather promising for HER UAVs multilabeling classification scenarios. The experimental results also report the significant improvements scored over a reference state-of-the-art method. Moreover, we believe that more investigations on the multiclass RBFNN output thresholding mechanism is worth addressing.

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